

Graphs based makeup transfer

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I. INTRODUCTION

Makeup transfer is a well-known problem in the field of Computer Vision. It aims to apply cosmetic styles such as lipstick, eyeshadow, and foundation from a reference image to a target face image while preserving the target's identity and skin tone. Inspired by recent advances in graph-based neural style transfer, this project proposes a Graph Neural Network (GNN)-based approach for makeup transfer. By representing faces as graphs with nodes corresponding to semantic regions (e.g., lips, eyes, skin) and edges encoding spatial and semantic relationships, we can selectively propagate makeup features across the graph.

GitHub: <https://github.com/sinhajiya/graphglow.git>

II. RELATED WORK

Our work builds upon the makeup transfer framework introduced by Li et al. ([1]). The paper uses histogram-based loss function which is useful for matching skin tone, but since it is global, it fails to capture local features properly. We use this idea of histogram for color matching and build our work on it.

III. FACE AS A GRAPH

We represent the face as a landmark graph, where each landmark serves as a node and edges encode spatial relationships. We construct two intra-face graphs by connecting each landmark to its k-nearest neighbors, capturing the local information (such as skin-tone) of the source and reference faces. For inter-face, we add edges linking corresponding regions (lips ↔ lips and eyes ↔ eyes) between faces so that color information can flow between faces through the edges. Intra face graph ensure that the transferred color is smooth. Fig. 1 shows one such example of the constructed graph. We only focus on 2 main regions where makeup is generally applied: lips



Fig. 1. Sample graph connection

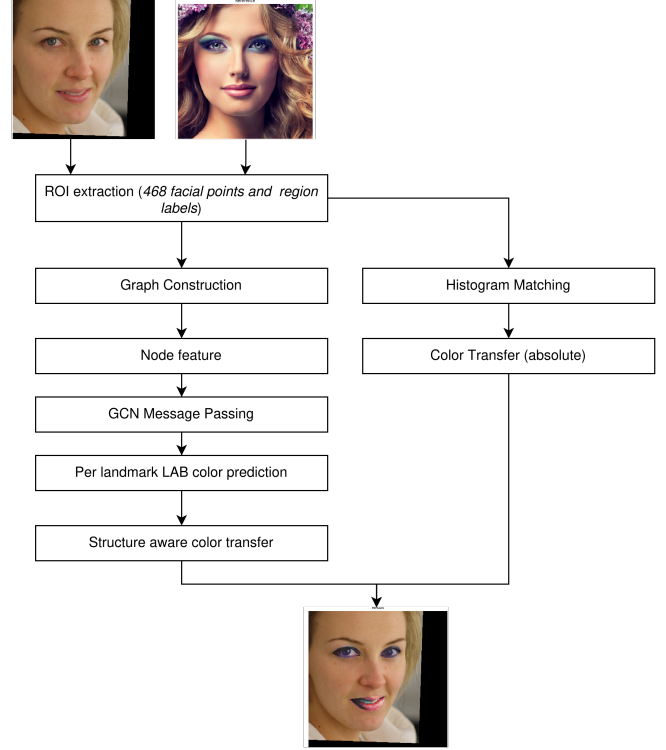


Fig. 2. Overall pipeline of our architecture

and eyes. Throughout this report, “source” refers to the non-makeup face (the face on which makeup is applied), and “reference” refers to the makeup face (the face from which the makeup style is transferred).

IV. MODELS AND ARCHITECTURES

A. Region-Aware Makeup Transfer using GCNs:

We model makeup transfer as a graph based color propagation problem. Facial landmarks provide a precise geometry of lips and eyes, enabling us to build a structure-aware graph on which a GCN learns how makeup color should flow across the boundaries. Region-wise histogram matching gives a deterministic alignment initially, while the GCN operates on the graphs to transfer the color smoothly. It is responsible for enforcing continuity, correcting local inconsistencies, and propagating region-specific style cues across the face. Fig. 2 shows the overall architecture.

a) Facial Region Graph Construction:

Using MediaPipe’s 468 landmarks, we extract only the region-specific landmarks for lips and eyes. These landmarks form the nodes of two graphs:

- Source face graph
- Reference face graph

We form the inter and intra face graphs as mentioned in Section III. We extracted the landmarks manually from mediapipe.

b) *Node Features:*

Each node is assigned a 9-dimensional feature vector extracted from its local pixel patch:

- Mean LAB (3 dims)
- Standard deviation LAB (3 dims)
- Mean RGB (3 dims)

These features capture the local color statistics that the GCN learns to adjust. Because the graph is sparse and semantically organized, these features are sufficient to drive coherent, structure-aligned makeup refinement.

c) *Two-Stage Color Transfer Pipeline:*

Color transfer is performed in two stages:

Histogram Matching: We first apply region-wise histogram matching to align the source region’s raw distribution to the reference. This step transfers global color but introduces minor discontinuities along region boundaries. It serves as a pre-alignment that sets the target color space for the GCN.

Region-aware smoothing via GCN: Fig. 3 shows the refinement stage. The GCN performs color propagation over the graph: each node aggregates information from its kNN neighbors, spreading color smoothly along true facial geometry.

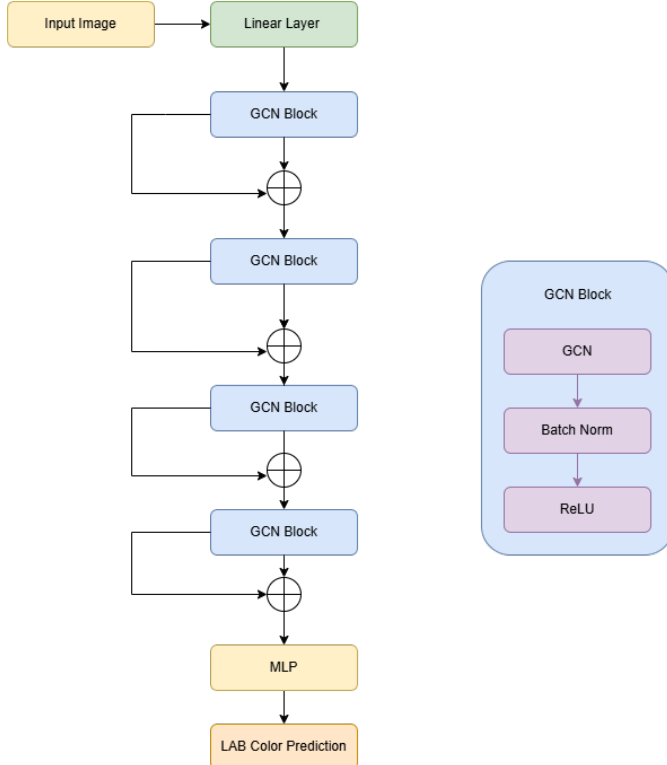


Fig. 3. GCN refinement architecture

Cross-face edges inject reference color signals directly into the corresponding source regions. Residual blocks prevent over-smoothing, allowing the network to preserve essential local variations while correcting histogram artifacts. This produces a learned, geometry-respecting refinement that simple blurring or mask-based filtering cannot reproduce.

d) *Final Reconstruction:* The GCN outputs refined LAB values for the source landmarks. These are propagated to the full region using convex-hull masks and cubic interpolation, producing a smooth per-region color field. A soft spatial mask blends the interpolated LAB field back into the histogram-matched image. The final result is a stable, graph-driven makeup transfer system that respects facial structure, avoids heavy generative models, and uses message passing as the central mechanism for style propagation.

V. DATASET

We evaluate our approach on a benchmark datasets. The MT dataset, first introduced with BeautyGAN [1], contains 3,834 female face images, including 1,115 non-makeup and 2,719 makeup examples.

VI. EXPERIMENTS AND RESULTS

We split the MT dataset into 75% training and 15% testing and trained the model for 100 epochs using the Adam optimizer with a cosine-annealing learning rate schedule.

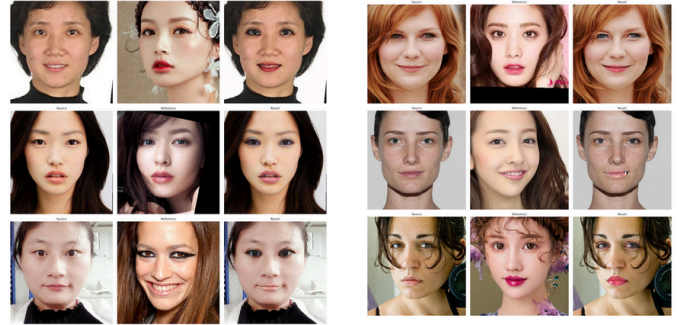


Fig. 4. Results of our model. Images shown as [source | reference | transferred].

A. User review

We evaluate the quality of our proposed model using a small user review. Ten random pairs of makeup and non-makeup image along with the transferred image were rated by 54 human raters on a scale of 1-10 (1=worst transfer, 10=best transfer) according to how well the makeup is transferred. We computed the mean score of the individual images by all the users shown in Fig. 5. The total mean score given by the users is **4.9/10** suggesting that our model has still a lot to learn,

TABLE I
LIMITATIONS OF OUR MODEL

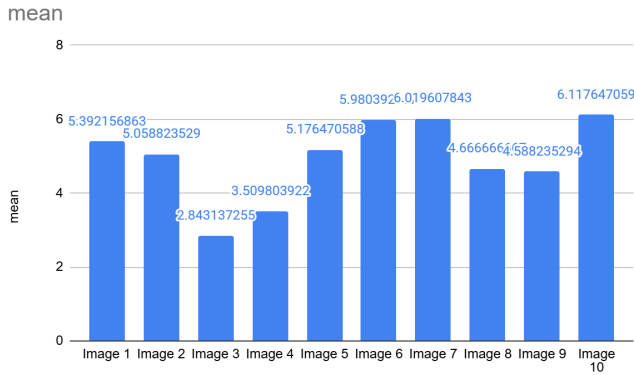
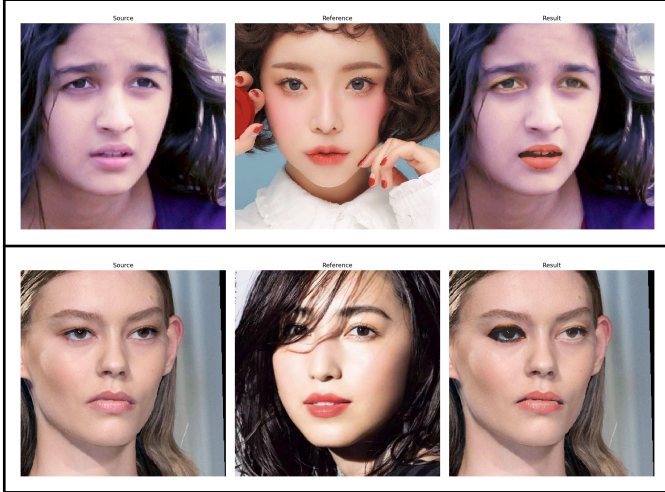


Fig. 5. User review

VII. LIMITATIONS

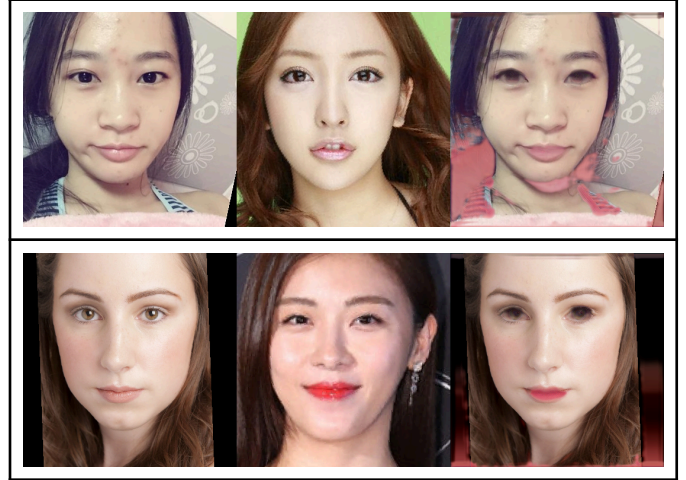
- Currently, our model doesn't focus on cheek regions where styles like - contouring and blushing is done. This part is challenging since using the same idea as above will lead to transfer of skin tone if landmarks are pre-defined: which is one of the main limitaion.
- When the reference contains occlusions such as hair overlapping the face, the model is not able to distinguish hair from facial regions and incorrectly blends those colors into the reference. Table I shows the visual of the limitations of our model.

VIII. MISCELLENOUS: OUR INITIAL IDEA

Our initial idea was to use an expressive GAT-based architecture, leading to the design of XFlow-GAT.

XFlow-GAT represents both faces as graphs and applies GATv2 layers to propagate color and texture across intra-face geometry and cross-face semantic links. While the approach is flexible and captures fine-grained correspondences, the full pipeline: message passing, landmark-wise refinement, rasterization, and a U-Net decoder proved unnecessarily complex for makeup transfer. In practice, the model hallucinated

TABLE II
WHY XFlow-GAT IDEA WAS DROPPED



unnatural colors, oversmoothed key regions, and failed to maintain global consistency. Table II shows an example of the output produced by XFlow-GAT. It shows overly smooth eyelids, distorted lip coloration, and color bleeding into the background demonstrate the instability of this approach when used for direct makeup transfer. These limitations indicated that GAT-based propagation was not well suited for this task so we shifted to simpler approach.

REFERENCES

- [1] T. Li *et al.*, "BeautyGAN: Instance-level Facial Makeup Transfer with Deep Generative Adversarial Network," in *Proceedings of the 26th ACM International Conference on Multimedia*, in MM '18. Seoul, Republic of Korea: Association for Computing Machinery, 2018, pp. 645–653. doi: 10.1145/3240508.3240618.